SECTION 6 — COMMUNICATIONS

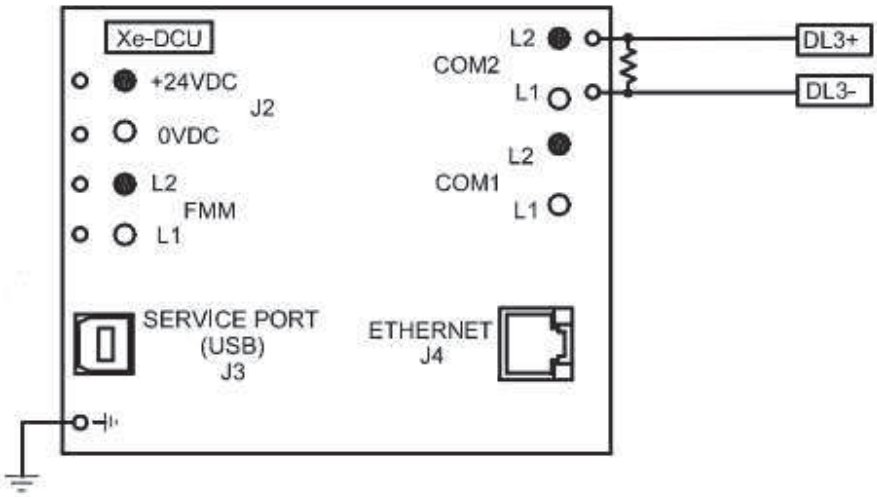
OVERVIEW

Connections to user systems (DCS, PLC, etc.) are done through the DCU. The DCU has both a RS-485 and Ethernet connection for this purpose. The RS-485 port supports Modbus RTU. The Ethernet supports Modbus TCP. The configuration of these ports is done through the DCU. RS-485 has configurable baud rate, stop bit, data bit, and parity. Ethernet has configuration IP addresses.

SUPPORTED PROTOCOLS

The DCU directly supports Modbus RTU and Modbus TCP. Other protocols may be supported through the use of additional converters. Refer to the manual for those devices for more information.

The RS-485 connection is made to terminal blocks in the panel. These will be labeled as DL3+ and DL3-. The Ethernet connection is made directly to the DCU. This is illustrated in the picture below:



Modbus architecture is designed as master and slave. The master (DCS, PLC, etc.) sends a message to a slave (Xe-145F). The slave only sends messages in response to a message from the master. Each slave has an address. This is configured on the DCU and labeled as Modbus. The screen below is found on the General Settings tab (see page 49)

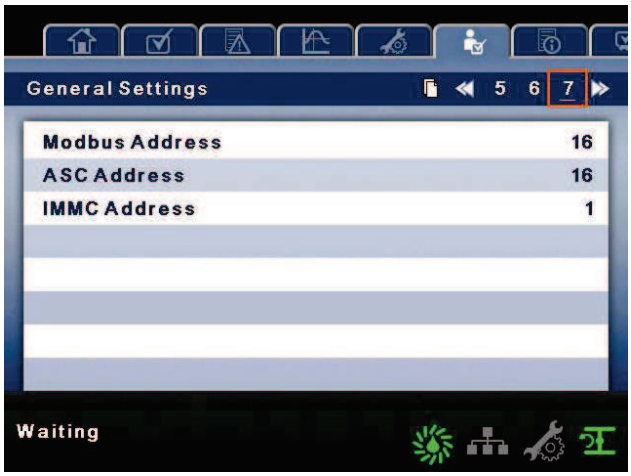


Figure 23: DCU Modbus Address Configuration

Modbus TCP also requires the IP address of the controller. This is also found and configured on the DCU. The screen below from the General Settings tab (see page 51) indicates how the controller IP's address is configured.

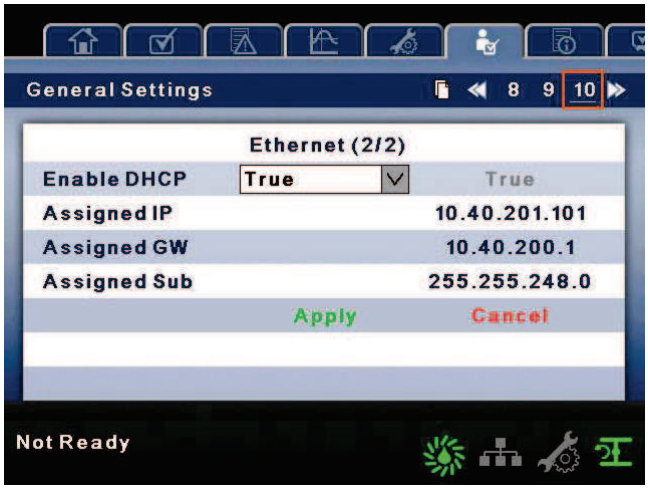


Figure 24: DCU Ethernet IP Information

The master may request multiple parameters in one message. The DCU has limitations on how many parameters it can process in a single message. The limitations are below:

Function Code	Description	Maximum Parameters
01	Read Coil Status	512 coils
02	Read Input Status	512 inputs
03	Read Holding Registers	64 registers
04	Read Input Registers	64 registers
05	Force Single Coil	1 coil
06	Preset Single Register	1 register
15	Force Multiple Coils	512 coils
16	Preset Multiple Registers	64 registers

Refer to hiip://www.modbus.org/specs.php for complete specifications for the Modbus protocol.

REGISTER ADDRESSES

Two addresses are shown for each parameter. The address to use is dependent on the PLC or software in the master system. Wonderware products, for example, use the absolute address.

The absolute address is truly a combination of function code and address. The relative address is an address only. The appropriate function code must be known when using relative addressing.

The compressor's schematics are necessary to find addresses in the following sections.

RELAY OUTPUTS AND COMPRESSOR STATUS

The compressor status and value of output relays may be read using Modbus function code 01 (Read Coil Status). The following table lists addresses:

Absolute Address (decimal)	Relative Address (decimal)	Coil Name
00187	0186	Digital Output, Channel 1 (J15-P7,8)
00188	0187	Digital Output, Channel 2 (J15-P5,6)
00189	0188	Digital Output, Channel 3 (J15-P3,4)
00190	0189	Digital Output, Channel 4 (J15-P1,2)
00191	0190	Digital Output, Channel 5 (J14-P7,8)
00192	0191	Digital Output, Channel 6 (J14-P5,6)
00193	0192	Digital Output, Channel 7 (J14-P3,4)
00194	0193	Digital Output, Channel 8 (J14-P1,2)
00195	0194	Digital Output, Channel 9 (J13-P7,8)
00196	0195	Digital Output, Channel 10 (J13-P5,6)
00197	0196	Digital Output, Channel 11 (J13-P3,4)
00198	0197	Digital Output, Channel 12 (J13-P1,2)
00199	0198	Digital Output, Channel 13 (J12-P7,8)
00200	0199	Digital Output, Channel 14 (J12-P5,6)
00201	0200	Digital Output, Channel 15 (J12-P3,4)
00202	0201	Digital Output, Channel 16 (J12-P1,2)
00203	0202	Compressor State - Waiting
00204	0203	Compressor State - Coasting
00205	0204	Compressor State - Starting
00206	0205	Compressor State - Not Ready
00207	0206	Compressor State – Ready
00208	0207	Compressor State - Surge Unload
00209	0208	Compressor State - Autodual Unload
00210	0209	Compressor State - Unloading
00211	0210	Compressor State - Unloaded
00212	0211	Compressor State - Min load
00213	0212	Compressor State - Max load
00214	0213	Compressor State - Loading
00215	0214	Compressor State - Loaded
00216	0215	Compressor State - Full Load
00218	0217	Any Compressor Trip
00219	0218	Any Compressor Warning

DIGITAL INPUTS

The status of digital inputs may be read by Modbus function code 02 (Read Input Status). The following table lists addresses:

Absolute Address (decimal)	Relative Address (decimal)	Input Name
10171	0170	Digital Input, Channel 1 (J4-P2)
10172	0171	Digital Input, Channel 2 (J4-P3)
10173	0172	Digital Input, Channel 3 (J4-P4)
10174	0173	Digital Input, Channel 4 (J4-P5)
10175	0174	Digital Input, Channel 5 (J4-P6)
10176	0175	Digital Input, Channel 6 (J4-P7)
10177	0176	Digital Input, Channel 7 (J4-P8)
10178	0177	Digital Input, Channel 8 (J4-P9)
10179	0178	Digital Input, Channel 9 (J5-P2)
10180	0179	Digital Input, Channel 10 (J5-P3)
10181	0180	Digital Input, Channel 11 (J5-P4)
10182	0181	Digital Input, Channel 12 (J5-P5)
10183	0182	Digital Input, Channel 13 (J5-P6)
10184	0183	Digital Input, Channel 14 (J5-P7)
10185	0184	Digital Input, Channel 15 (J5-P8)
10186	0185	Digital Input, Channel 16 (J5-P9)

ANALOG INPUTS

The value of analog inputs may be read using function 04 (Read Input Registers). The Xe controller supports two methods for determining the value for each holding register. (This also applies to Input Registers.) The first method uses two 16-bit integers to represent the integer and fraction part of the value. The second method uses one 32-bit IEEE floating point number. The following table lists addresses:

Input Register Name	Signed 16-Bit Integer		Unsigned 16-Bit Fraction		Signed IEEE 32-Bit Float	
	Absolute Address (decimal)	Relative Address (decimal)	Absolute Address (decimal)	Relative Address (decimal)	Absolute Address (decimal)	Relative Address (decimal)
Analog Input, Channel 1 (J2-P1,3)	30003	0002	30004	0003	33003	3002
Analog Input, Channel 2 (J2-P5,7)	30005	0004	30006	0005	33005	3004
Analog Input, Channel 3 (J1-P1)	30007	0006	30008	0007	33007	3006
Analog Input, Channel 4 (J1-P4)	30009	0008	30010	0009	33009	3008
Analog Input, Channel 5 (J1-P5)	30011	0010	30012	0011	33011	3010
Analog Input, Channel 6 (J1-P8)	30013	0012	30014	0013	33013	3012
Analog Input, Channel 7 (J1-P9)	30015	0014	30016	0015	33015	3014
Analog Input, Channel 8 (J1-P12)	30017	0016	30018	0017	33017	3015
Analog Input, Channel 9 (J1-P13)	30019	0018	30020	0019	33019	3018
Analog Input, Channel 10 (J1-P16)	30021	0020	30022	0021	33021	3020
Analog Input, Channel 11 (J1-P17)	30023	0022	30024	0023	33023	3022
Analog Input, Channel 12 (J1-P20)	30025	0024	30026	0025	33025	3024
Analog Input, Channel 13 (J1-P21)	30027	0026	30028	0027	33027	3026
Analog Input, Channel 14 (J1-P24)	30029	0028	30030	0029	33029	3028
Analog Input, Channel 15 (J1-P25)	30031	0030	30032	0031	33031	3030
Analog Input, Channel 16 (J1-P28)	30033	0032	30034	0033	33033	3032
Analog Input, Channel 17 (J1-P29)	30035	0034	30036	0035	33035	3034
Analog Input, Channel 18 (J1-P32)	30037	0036	30038	0037	33037	3036
Analog Input, Channel 19 (J1-P33)	30039	0038	30040	0039	33039	3038
Analog Input, Channel 20 (J1-P36)	30041	0040	30042	0041	33041	3040
Analog Input, Channel 21 (J1-P37)	30043	0042	30044	0043	33043	3042
Analog Input, Channel 22 (J1-P40)	30045	0044	30046	0045	33045	3044
Analog Input, Channel 23 (J1-P41)	30047	0046	30048	0047	33047	3046
CT Input (J9-P1,2)	30049	0048	30050	0049	33049	3048

ANALOG OUTPUTS, SETTINGS, AND CALCULATED DATA

The value of analog outputs, and various settings may be read using Modbus function code 03 (Read Holding Registers), and set using either function code 06 (Preset Single Register) or function code 16 (Preset Multiple Registers). The following table lists addresses. This is supported by the following functions:

Holding Register Name - Read/Write	Signed 16 Bit Exponent		Unsigned 16 Bit Fraction		Signed IEEE 32-Bit Float	
	Absolute Address (decimal)	Relative Address (decimal)	Absolute Address (decimal)	Relative Address (decimal)	Absolute Address (decimal)	Relative Address (decimal)
Analog Output, Channel 1 (J3-P1,3)	40053	0052	40054	0053	43053	3052
Analog Output, Channel 2 (J3-P4,6)	40055	0054	40056	0055	43055	3054
Analog Input, Channel 1 (J2-P1,3) - High Trip Setpoint	40061	0060	40062	0061	43061	3060
Analog Input, Channel 1 (J2-P1,3) - High Warning Setpoint	40063	0062	40064	0063	43063	3062
Analog Input, Channel 1 (J2-P1,3) - Low Warning Setpoint	40065	0064	40066	0065	43065	3064
Analog Input, Channel 1 (J2-P1,3) - Low Trip Setpoint	40067	0066	40068	0067	43067	3066
Analog Input, Channel 2 (J2-P5,7) - High Trip Setpoint	40069	0068	40070	0069	43069	3068
Analog Input, Channel 2 (J2-P5,7) - High Warning Setpoint	40071	0070	40072	0071	43071	3070
Analog Input, Channel 2 (J2-P5,7) - Low Warning Setpoint	40073	0072	40074	0073	43073	3072
Analog Input, Channel 2 (J2-P5,7) - Low Trip Setpoint	40075	0074	40076	0075	43075	3074
Analog Input, Channel 3 (J1-P1) - High Trip Setpoint	40077	0076	40078	0077	43077	3076
Analog Input, Channel 3 (J1-P1) - High Warning Setpoint	40079	0078	40080	0079	43079	3078
Analog Input, Channel 3 (J1-P1) - Low Warning Setpoint	40081	0080	40082	0081	43081	3080
Analog Input, Channel 3 (J1-P1) - Low Trip Setpoint	40083	0082	40084	0083	43083	3082
Analog Input, Channel 4 (J1-P4) - High Trip Setpoint	40085	0084	40086	0085	43085	3084
Analog Input, Channel 4 (J1-P4) - High Warning Setpoint	40087	0086	40088	0087	43087	3086
Analog Input, Channel 4 (J1-P4) - Low Warning Setpoint	40089	0088	40090	0089	43089	3088
Analog Input, Channel 4 (J1-P4) - Low Trip Setpoint	40091	0090	40092	0091	43091	3090
Analog Input, Channel 5 (J1-P5) - High Trip Setpoint	40093	0092	40094	0093	43093	3092
Analog Input, Channel 5 (J1-P5) - High Warning Setpoint	40095	0094	40096	0095	43095	3094
Analog Input, Channel 5 (J1-P5) - Low Warning Setpoint	40097	0096	40098	0097	43097	3096
Analog Input, Channel 5 (J1-P5) - Low Trip Setpoint	40099	0098	40100	0099	43099	3098
Analog Input, Channel 6 (J1-P8) - High Trip Setpoint	40101	0100	40102	0101	43101	3100
Analog Input, Channel 6 (J1-P8) - High Warning Setpoint	40103	0102	40104	0103	43103	3102
Analog Input, Channel 6 (J1-P8) - Low Warning Setpoint	40105	0104	40106	0105	43105	3104
Analog Input, Channel 6 (J1-P8) - Low Trip Setpoint	40107	0106	40108	0107	43107	3106
Analog Input, Channel 7 (J1-P9) - High Trip Setpoint	40109	0108	40110	0109	43109	3108
Analog Input, Channel 7 (J1-P9) - High Warning Setpoint	40111	0110	40112	0111	43111	3110
Analog Input, Channel 7 (J1-P9) - Low Warning Setpoint	40113	0112	40114	0113	43113	3112
Analog Input, Channel 7 (J1-P9) - Low Trip Setpoint	40115	0114	40116	0115	43115	3114
Analog Input, Channel 8 (J1-P12) - High Trip Setpoint	40117	0116	40118	0117	43117	3116
Analog Input, Channel 8 (J1-P12) - High Warning Setpoint	40119	0118	40120	0119	43119	3118
Analog Input, Channel 8 (J1-P12) - Low Warning Setpoint	40121	0120	40122	0121	43121	3120
Analog Input, Channel 8 (J1-P12) - Low Trip Setpoint	40123	0122	40124	0123	43123	3122
Analog Input, Channel 9 (J1-P13) - High Trip Setpoint	40125	0124	40126	0125	43125	3124
Analog Input, Channel 9 (J1-P13) - High Warning Setpoint	40127	0126	40128	0127	43127	3126
Analog Input, Channel 9 (J1-P13) - Low Warning Setpoint	40129	0128	40130	0129	43129	3128
Analog Input, Channel 9 (J1-P13) - Low Trip Setpoint	40131	0130	40132	0131	43131	3130
Analog Input, Channel 10 (J1-P16) - High Trip Setpoint	40133	0132	40134	0133	43133	3132
Analog Input, Channel 10 (J1-P16) - High Warning Setpoint	40135	0134	40136	0135	43135	3134
Analog Input, Channel 10 (J1-P16) - Low Warning Setpoint	40137	0136	40138	0137	43137	3136
Analog Input, Channel 10 (J1-P16) - Low Trip Setpoint	40139	0138	40140	0139	43139	3138
Analog Input, Channel 11 (J1-P17) - High Trip Setpoint	40141	0140	40142	0141	43141	3140
Analog Input, Channel 11 (J1-P17) - High Warning Setpoint	40143	0142	40144	0143	43143	3142
Analog Input, Channel 11 (J1-P17) - Low Warning Setpoint	40145	0144	40146	0145	43145	3144
Analog Input, Channel 11 (J1-P17) - Low Trip Setpoint	40147	0146	40148	0147	43147	3146
Analog Input, Channel 12 (J1-P20) - High Trip Setpoint	40149	0148	40150	0149	43149	3148
Analog Input, Channel 12 (J1-P20) - High Warning Setpoint	40151	0150	40152	0151	43151	3150
Analog Input, Channel 12 (J1-P20) - Low Warning Setpoint	40153	0152	40154	0153	43153	3152
Analog Input, Channel 12 (J1-P20) - Low Trip Setpoint	40155	0154	40156	0155	43155	3154
Analog Input, Channel 13 (J1-P21) - High Trip Setpoint	40157	0156	40158	0157	43157	3156
Analog Input, Channel 13 (J1-P21) - High Warning Setpoint	40159	0158	40160	0159	43159	3158
Analog Input, Channel 13 (J1-P21) - Low Warning Setpoint	40161	0160	40162	0161	43161	3160
Analog Input, Channel 13 (J1-P21) - Low Trip Setpoint	40163	0162	40164	0163	43163	3162

Holding Register Name - Read/Write	Signed 16 Bit Exponent		Unsigned 16 Bit Fraction		Signed IEEE 32-Bit Float	
	Absolute Address (decimal)	Relative Address (decimal)	Absolute Address (decimal)	Relative Address (decimal)	Absolute Address (decimal)	Relative Address (decimal)
Analog Input, Channel 14 (J1-P24) - High Trip Setpoint	40165	0164	40166	0165	43165	3164
Analog Input, Channel 14 (J1-P24) - High Warning Setpoint	40167	0166	40168	0167	43167	3166
Analog Input, Channel 14 (J1-P24) - Low Warning Setpoint	40169	0168	40170	0169	43169	3168
Analog Input, Channel 14 (J1-P24) - Low Trip Setpoint	40171	0170	40172	0171	43171	3170
Analog Input, Channel 15 (J1-P25) - High Trip Setpoint	40173	0172	40174	0173	43173	3172
Analog Input, Channel 15 (J1-P25) - High Warning Setpoint	40175	0174	40176	0175	43175	3174
Analog Input, Channel 15 (J1-P25) - Low Warning Setpoint	40177	0176	40178	0177	43177	3176
Analog Input, Channel 15 (J1-P25) - Low Trip Setpoint	40179	0178	40180	0179	43179	3178
Analog Input, Channel 16 (J1-P28) - High Trip Setpoint	40181	0180	40182	0181	43181	3180
Analog Input, Channel 16 (J1-P28) - High Warning Setpoint	40183	0182	40184	0183	43183	3182
Analog Input, Channel 16 (J1-P28) - Low Warning Setpoint	40185	0184	40186	0185	43185	3184
Analog Input, Channel 16 (J1-P28) - Low Trip Setpoint	40187	0186	40188	0187	43187	3186
Analog Input, Channel 17 (J1-P29) - High Trip Setpoint	40189	0188	40190	0189	43189	3188
Analog Input, Channel 17 (J1-P29) - High Warning Setpoint	40191	0190	40192	0191	43191	3190
Analog Input, Channel 17 (J1-P29) - Low Warning Setpoint	40193	0192	40194	0193	43193	3192
Analog Input, Channel 17 (J1-P29) - Low Trip Setpoint	40195	0194	40196	0195	43195	3194
Analog Input, Channel 18 (J1-P32) - High Trip Setpoint	40197	0196	40198	0197	43197	3196
Analog Input, Channel 18 (J1-P32) - High Warning Setpoint	40199	0198	40200	0199	43199	3198
Analog Input, Channel 18 (J1-P32) - Low Warning Setpoint	40201	0200	40202	0201	43201	3200
Analog Input, Channel 18 (J1-P32) - Low Trip Setpoint	40203	0202	40204	0203	43203	3202
Analog Input, Channel 19 (J1-P33) - High Trip Setpoint	40205	0204	40206	0205	43205	3204
Analog Input, Channel 19 (J1-P33) - High Warning Setpoint	40207	0206	40208	0207	43207	3206
Analog Input, Channel 19 (J1-P33) - Low Warning Setpoint	40209	0208	40210	0209	43209	3208
Analog Input, Channel 19 (J1-P33) - Low Trip Setpoint	40211	0210	40212	0211	43211	3210
Analog Input, Channel 20 (J1-P36) - High Trip Setpoint	40213	0212	40214	0213	43213	3212
Analog Input, Channel 20 (J1-P36) - High Warning Setpoint	40215	0214	40216	0215	43215	3214
Analog Input, Channel 20 (J1-P36) - Low Warning Setpoint	40217	0216	40218	0217	43217	3216
Analog Input, Channel 20 (J1-P36) - Low Trip Setpoint	40219	0218	40220	0219	43219	3218
Analog Input, Channel 21 (J1-P37) - High Trip Setpoint	40221	0220	40222	0221	43221	3220
Analog Input, Channel 21 (J1-P37) - High Warning Setpoint	40223	0222	40224	0223	43223	3222
Analog Input, Channel 21 (J1-P37) - Low Warning Setpoint	40225	0224	40226	0225	43225	3224
Analog Input, Channel 21 (J1-P37) - Low Trip Setpoint	40227	0226	40228	0227	43227	3226
Analog Input, Channel 22 (J1-P40) - High Trip Setpoint	40229	0228	40230	0229	43229	3228
Analog Input, Channel 22 (J1-P40) - High Warning Setpoint	40231	0230	40232	0231	43231	3230
Analog Input, Channel 22 (J1-P40) - Low Warning Setpoint	40233	0232	40234	0233	43233	3232
Analog Input, Channel 22 (J1-P40) - Low Trip Setpoint	40235	0234	40236	0235	43235	3234
Analog Input, Channel 23 (J1-P41) - High Trip Setpoint	40237	0236	40238	0237	43237	3236
Analog Input, Channel 23 (J1-P41) - High Warning Setpoint	40239	0238	40240	0239	43239	3238
Analog Input, Channel 23 (J1-P41) - Low Warning Setpoint	40241	0240	40242	0241	43241	3240
Analog Input, Channel 23 (J1-P41) - Low Trip Setpoint	40243	0242	40244	0243	43243	3242
Motor Current	40267	0266	40268	0267	43267	3266
Target Pressure	40269	0268	40270	0269	43269	3268
Surge Point	40271	0270	40272	0271	43271	3270
Motor Full Load Amps	40273	0272	40274	0273	43273	3272
Autodual Reload Percent	40275	0274	40276	0275	43275	3274
Autodual Unload Point	40277	0276	40278	0277	43277	3276
Autodual Unload Timer	40279	0278	40280	0279	43279	3278
Pressure Setpoint Ramp Rate	40281	0280	40282	0281	43281	3280
Inlet Valve Unload Position	40283	0282	40284	0283	43283	3282
Start Timer	40285	0284	40286	0285	43285	3284
CT Ratio	40287	0286	40288	0287	43287	3286
Power On Hours	40297	0296	40298	0297	43297	3296
Running Hours	40299	0298	40300	0299	43299	3298
Loaded Hours	40301	0300	40302	0301	43301	3300
Number of Starts	40303	0302	40304	0303	43303	3302
Inlet Valve, MaxLoad, Proportional Constant	40313	0312	40314	0313	43313	3312
Inlet Valve, MaxLoad, Integral Constant	40315	0314	40316	0315	43315	3314
Inlet Valve, MaxLoad, Derivative Constant	40317	0316	40318	0317	43317	3316
Inlet Valve, MinLoad, Proportional Constant	40319	0318	40320	0319	43319	3318
Inlet Valve, MinLoad, Integral Constant	40321	0320	40322	0321	43321	3320
Inlet Valve, MinLoad, Derivative Constant	40323	0322	40324	0323	43323	3322
Inlet Valve, Pressure, Proportional Constant	40325	0324	40326	0325	43325	3324
Inlet Valve, Pressure, Integral Constant	40327	0326	40328	0327	43327	3326
Inlet Valve, Pressure, Derivative Constant	40329	0328	40330	0329	43329	3328
Bypass Valve, Pressure, Proportional Constant	40331	0330	40332	0331	43331	3330
Bypass Valve, Pressure, Integral Constant	40333	0332	40334	0333	43333	3332
Bypass Valve, Pressure, Derivative Constant	40335	0334	40336	0335	43335	3334
Compressor Control Mode; 1=Modulate, 2=Autodual	40339	0338	40340	0339	43339	3338
NOTE: (J1-P1) is interpreted as Connector J1, Pin 1						

REMOTE START, STOP, LOAD, UNLOAD

Remote commands are supported by Modbus function codes 05 (Force Single Coil) and 15 (Force Multiple Coils).

IMPORTANT

For all of the following Remote Coils, the compressor's REMOTE COMMUNICATIONS DISABLED/ENABLED check box must be checked (see page 34) for these commands to execute. When DISABLED, the Xe-145F ignores (there is no exception response) these coils being forced ON or OFF.

Absolute Address (decimal)	Relative Address (decimal)	Coil Name - Write Only
00221	0220	Remote Horn Silence (Acknowledge)
00222	0221	Remote Reset
00223	0222	Remote Load
00224	0223	Remote Unload
00225	0224	Remote Start
00226	0225	Remote Stop

EXCEPTION CODES

The controller supports the following exception codes:

Name	Code	Description
Illegal Function	01	The function code received in the query is not an allowable action for the slave. This exception code happens when: (1) the function code is other than 1, 2, 3, 4, 5, 6, 15 or 16 (2) a message has the incorrect number of bytes for the function specified
Illegal Data Address	02	The data address received in the query is not an allowable address for the slave. This exception code happens when: (1) the address is not programmed (2) the address is outside of the ranges (a) 00001-00512 for coils (b) 10001-10512 for discrete inputs (c) 30001-31024 for integer and fractional analog inputs (d) 33001-34024 for floating point analog inputs (e) 40001-41024 for integer and fractional input registers (f) 43001-44024 for floating point analog input registers
Illegal Data Value	03	A value contained in the query data field is not an allowable value for the slave. This exception code happens when: (1) the number of coils, discrete inputs, registers or analog inputs is equal to zero (2) request for more than the maximum number of parameters (3) the force single coil command, Function 05, is issued and the value is other than FF00 or 0000 (4) the force multiple coil command, Function 15, is issued and the number of bytes does not equal the number of bits to set (5) the preset single register command, Function 6, or preset multiple registers commands, Function 16, is issued and the starting address is not even, or the number of registers specified does not correspond to the number of bytes in the message, or the integer part of the number is outside the range –32768 to +32767, or the fractional part of the number is outside of the range 0-9999, or the value is not a valid 32 bit floating point number

INSTALLATION GUIDELINES

Connections are made to the controller using RS485 or Ethernet. Either can work successfully if the following installation guidelines are followed:

RS-485

RS-485 is a proven method for communications in the industrial environment.

- The maximum distance from the first device on the network to the last device is 1219 m (4000 ft). This distance is in electrical feet which should include vertical wire runs as well as any other routing.
- The maximum number of devices on a single network is 32.
- Wire can make or break an installation. There are many variations of wire. The recommended wire is Belden 9841. Other kinds of wire may work. This wire has proven to be reliable over a great distance in a number of environments. Features of this wire include
- Twisted pair
- 24 AWG
- 120 ohm impedance
- Tape shield providing 100% coverage
- Braided shield providing 90% coverage
- Wide temperature range
- Approved by multiple agencies (CE, UL, etc)
- Avoid routing the wire parallel to higher voltage cables. High voltage cables should be crossed perpendicular when necessary.
- Install two 120 ohm ½ watt resistors. One at each end of the network.
- Connect the shield drain wires together and ground in one place. This will prevent ground loops. There may be a situation where multiple ground connections on the shield is necessary if there is a particularly noisy environment and grounds are at the same potential.
- Connect to terminal blocks in the panel.
- Variations exist in labeling of the wires. IR uses L1 and L2. Please note the following:
    - L1 => B => +
    - L2 => A => -

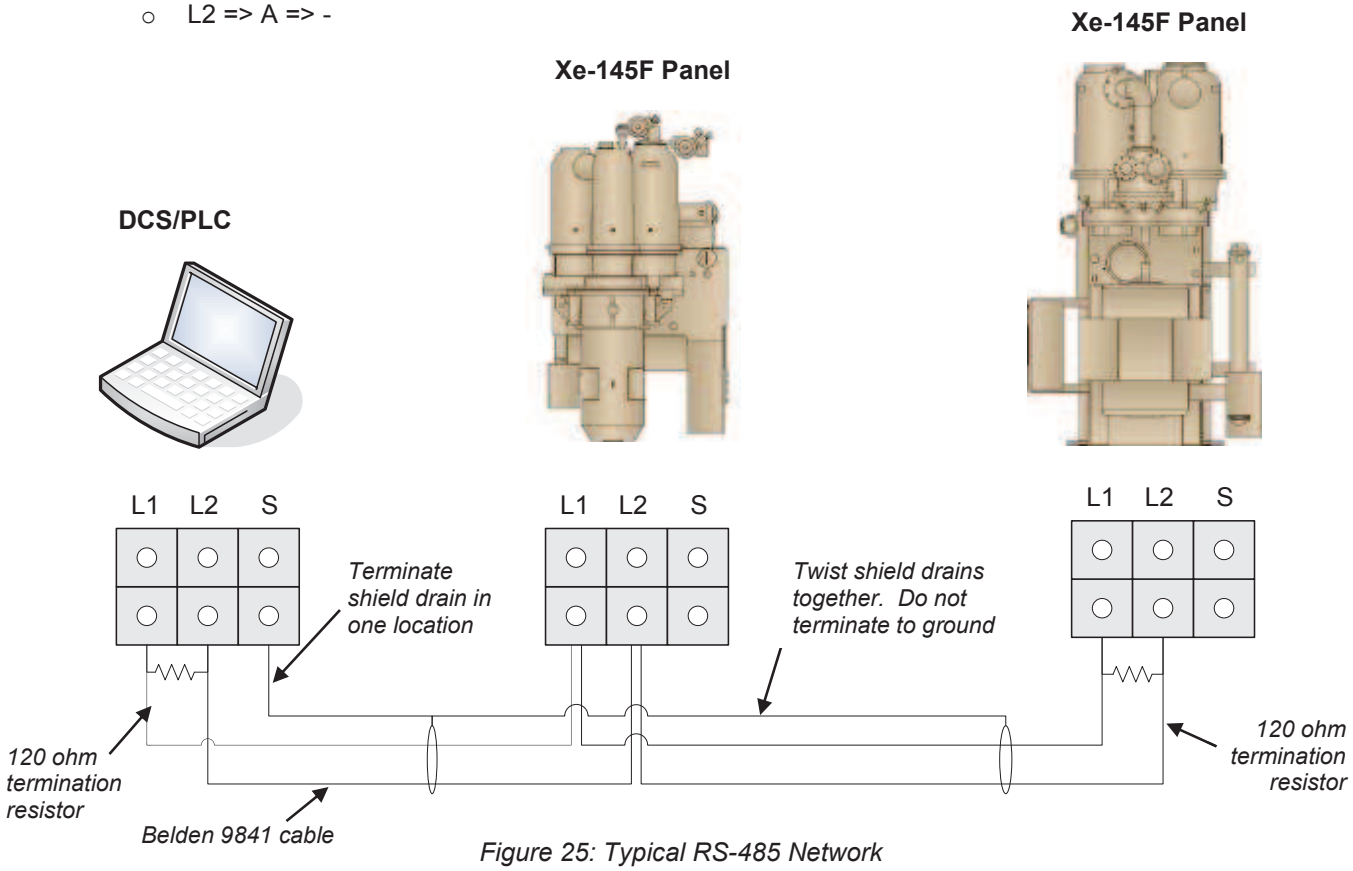


Figure 25: Typical RS-485 Network

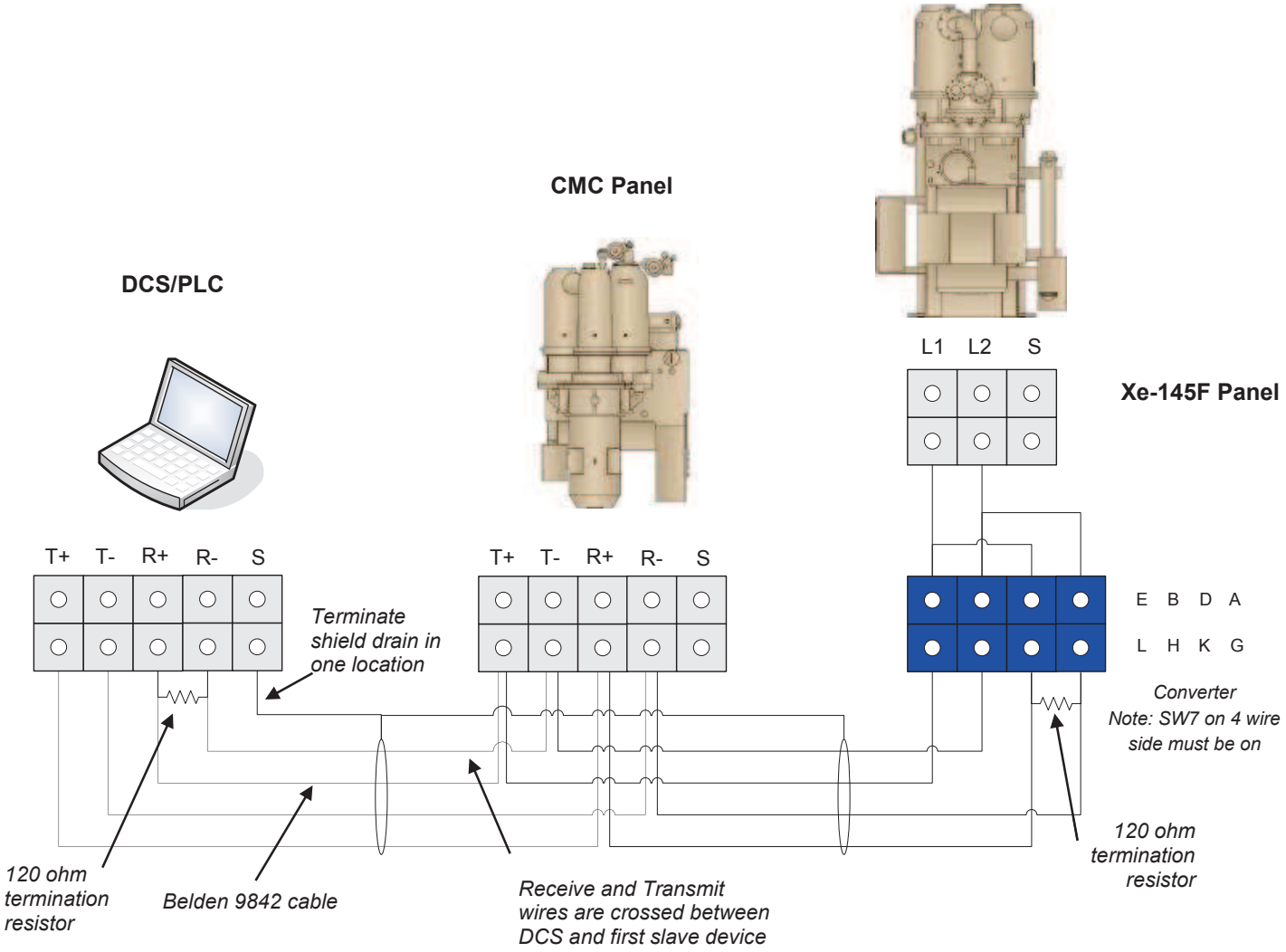


Figure 26: RS-422 Network



Figure 27: RS-485/RS-422 Converter (IR P/N 22475917)

ETHERNET

Follow normal Ethernet guidelines during installation.

- The maximum distance from point to point is 100 m (328 ft).
- Connect directly to the DCU.
- Connection directly to a PC (vs. to wall jack or network switch) requires a cross-over cable. A cross-over cable inverses the transmit and receive lines so two devices can directly communicate to one another.
- Use Cat 5 or Cat 5e cable.
- Note that the Ethernet connection will timeout after 30 s of inactivity.



Cross Over Cable Must Be Used
Note: Device must be on the same subnet

Figure 28: Direct Connection

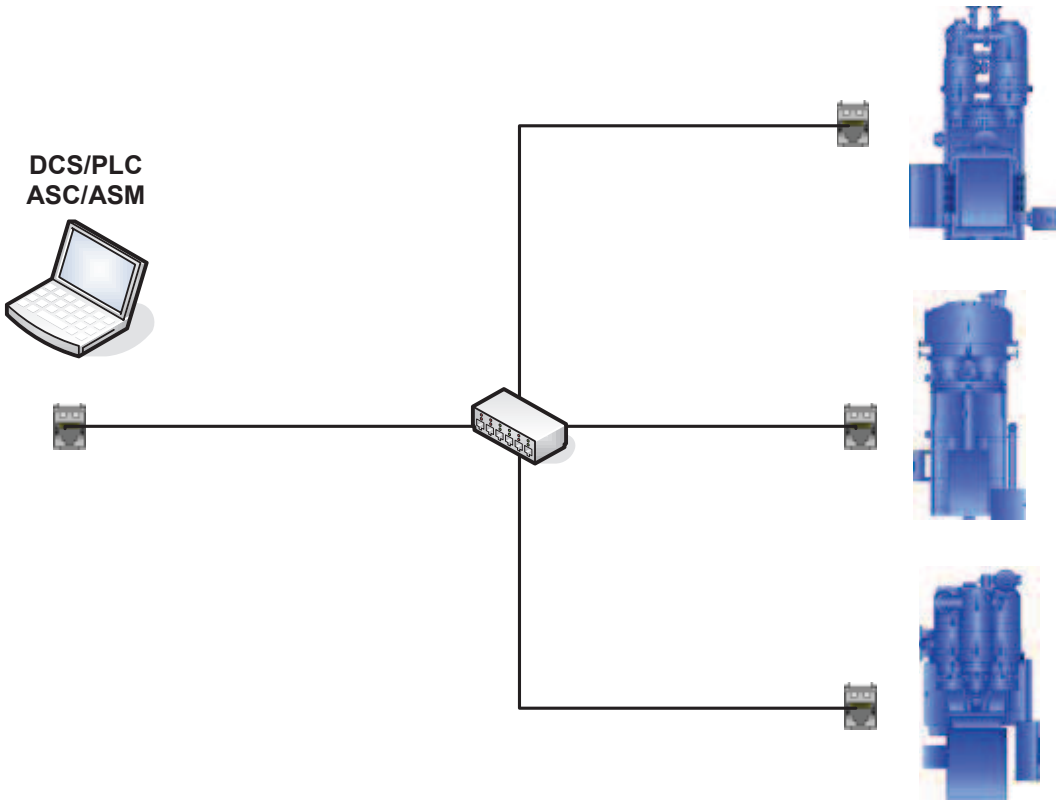


Figure 29: Multiple Ethernet Connections

GETTING ADDRESS INFO

Example 1: Read Analog Input Value

The schematics are needed to obtain address information. The pictures below are an excerpt from schematics:

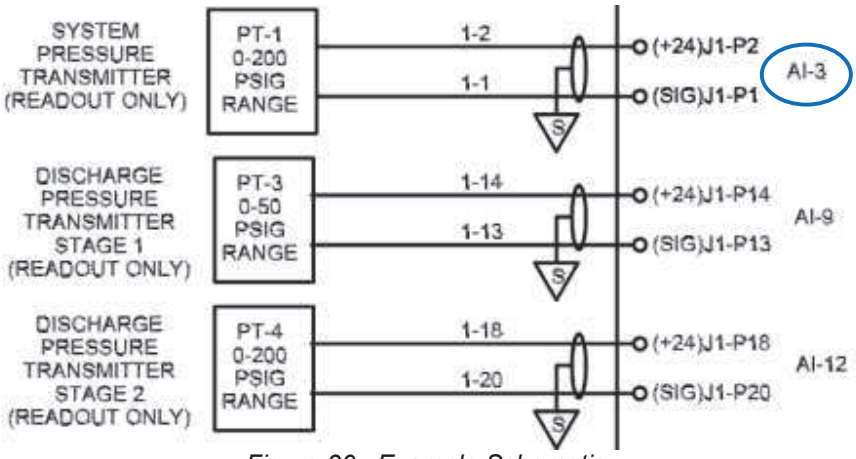


Figure 30: Example Schematic

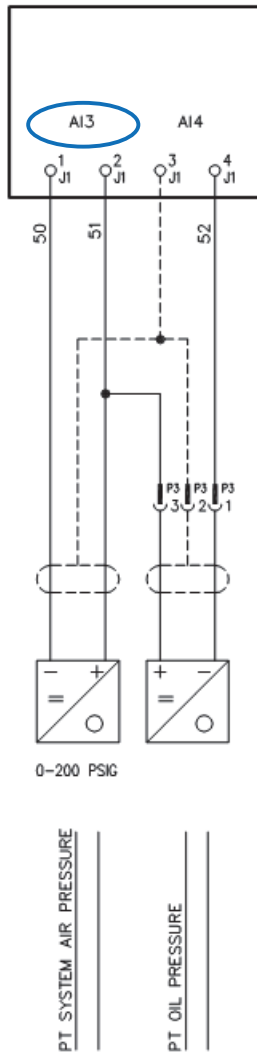


Figure 31: Example Schematic

The pressure transducer for System Pressure is shown above. It is wired to the controller. The channel is identified as AI-3. Use the table on page 92 to find the address for analog input 3. The table shows analog input 3 as registers 30007, 30008 or 33007. Suppose System Pressure is 93.2 on the screen. The values of the registers will be:

30007 = 93	integer portion (16 bit) of the number
30008 = 2101	decimal portion (16 bit) of the number
33007 = 93.2101	complete 32 bit number

The transmitted message to the controller is for the 30007 register is:
01 04 00 06 00 01 D1 CB

01	Modbus address of device. This address is configured on the screen.
04	Modbus function code for reading input registers.
00 06	Address 6. This is the relative address. The relative address is what is sent to the controller. The absolute address is used by systems such as Wonderware. The absolute address allows Wonderware to automatically determine the function code to use.
00 01	Total number of registers to get data
D1 CB	CRC. This value is calculated from the message. It is used to verify messages are correct. The message is ignored by the receiver if the CRC and message do not match.

The reply from the controller is:
01 04 02 00 5D 78 C9

01	Modbus address (echo of transmitted message if successful)
04	Function code for reading input registers (echo of message if successful)
02	Indicates that two bytes of data follows
00 5D	Two bytes of data. 5D converted from hex to decimal is 93
78 C9	This value is calculated from the message. It is used to verify messages are correct.

Example 2: Read Analog Output Values

The schematics are needed to obtain address information. The pictures below are an excerpt from schematics:

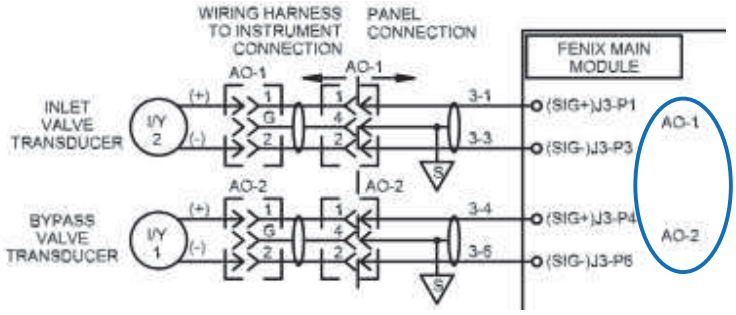


Figure 32: Example Schematic

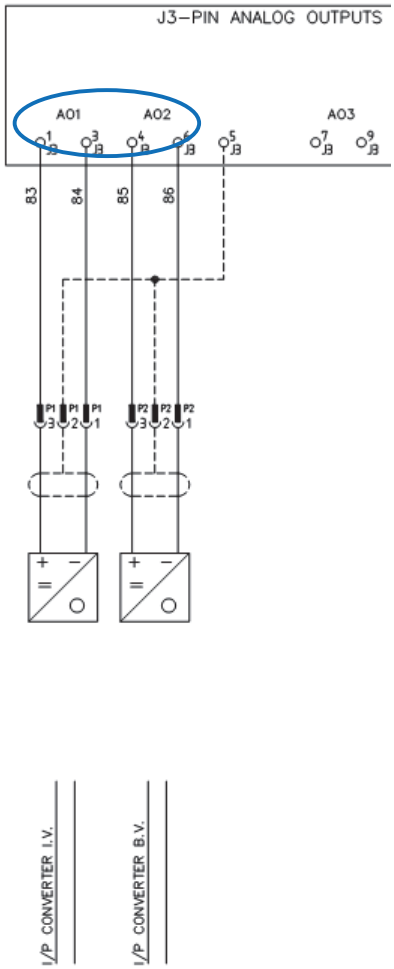


Figure 33: Example Schematic

The valve positioners are shown above. The channels are identified as analog output 1 and analog output 2. Use the table on page 93 to find the addresses for these channels. The table shows analog output 1 as 40053 and analog output 2 as 40055.

The example below collects both registers in one message. More data can be transferred from the controller at a quicker rate if fewer messages are used. Sending one message for two registers is much quicker than sending two different messages. The intent is to request 40053 – 40055.

The transmitted message to the controller is for the 40053 register (and two additional registers) is:
01 03 00 34 00 03 44 05

01	Modbus address of device. This address is configured on the screen.
03	Modbus function code for reading holding registers.
00 34	Address 34 hex is equivalent to 52 in decimal. This is the relative address. The relative address is what is sent to the controller. The absolute address is used by systems such as Wonderware. The absolute address allows Wonderware to automatically determine the function code to use.
00 03	Total number of registers to get data
44 05	CRC. This value is calculated from the message. It is used to verify messages are correct. The message is ignored by the receiver if the CRC and message do not match.

The reply from the controller is:
01 03 06 00 00 00 00 00 64 20 9E

01 Modbus address (echo of transmitted message if successful)
03 Function code for reading input registers (echo of message if successful)
06 Indicates that a total six bytes of data follows
00 00 Two bytes of data. The value of the first register is 0.
00 00 Two bytes of data. The value of the second register is 0.
00 64 Two bytes of data. The value of the third register is 64 hex or 100 decimal.
20 9E This value is calculated from the message. It is used to verify messages are correct.

This means 40053 = 0, 40054 = 0, 40055 = 100. The inlet valve is 0% open and the bypass is 100% open.

Example 3: Read Digital Inputs Values

The schematics are needed to obtain address information. The picture below is an excerpt from schematics:

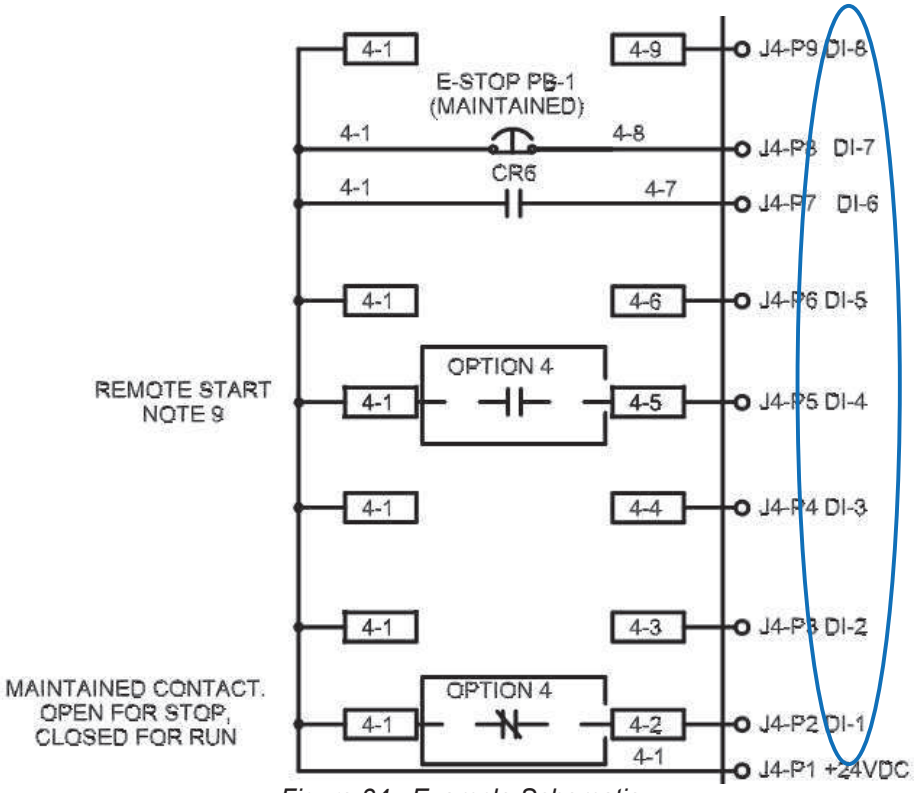


Figure 34: Example Schematic

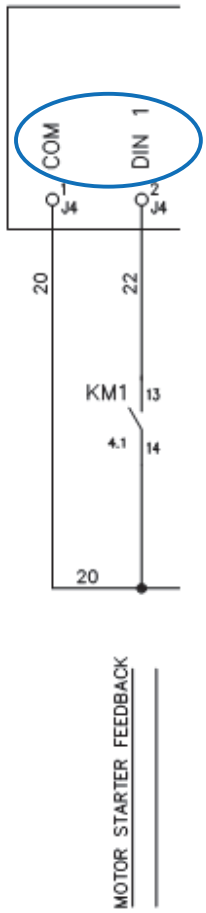


Figure 35: Example Schematic

The channels are identified as digital inputs 1 through 8. Use the table on page 91 to find the addresses for these channels. The table shows digital input 1 as 10171.

The example below collects all registers in one message. More data can be transferred from the controller at a quicker rate if fewer messages are used. Sending one message for two registers is much quicker than sending two different messages. The intent is to request 10171 – 10178.
The transmitted message to the controller is for the 10171 register (and eight additional registers) is:
01 02 00 AA 00 08 59 EC

01	Modbus address of device. This address is configured on the screen.
02	Modbus function code for reading input status.
00 AA	Address AA hex is equivalent to 170 in decimal. This is the relative address. The relative address is what is sent to the controller. The absolute address is used by systems such as Wonderware. The absolute address allows Wonderware to automatically determine the function code to use.
00 08	Total number of registers to get data
59 EC	CRC. This value is calculated from the message. It is used to verify messages are correct. The message is ignored by the receiver if the CRC and message do not match.

The reply from the controller is:
01 02 01 88 A1 EE

01	Modbus address (echo of transmitted message if successful)
02	Function code for reading input status (echo of message if successful)
01	Indicates that a total one byte of data follows
88	Bytes of data. 88 hex is converted to 1000 1000 in binary. The register data starts from right to left.
A1 EE	This value is calculated from the message. It is used to verify messages are correct.

This means:
10171 Remote Stop = 0 (from left)
10172 Not used = 0
10173 Not used = 0
10174 Remote Start = 1
10175 Not used = 0
10176 Not used= 0
10177 CR6 = 0
10178 E-Stop = 1. Note E-Stop is normally closed therefore this indicates that E-Stop is not pressed.

Example 4: Read Digital Outputs

The schematics are needed to obtain address information. The picture below is an excerpt from schematics:

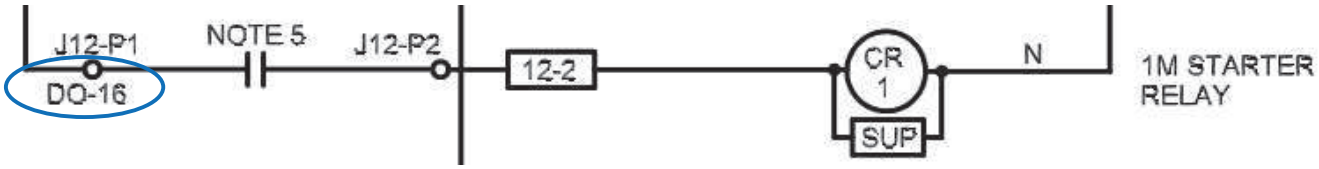


Figure 36: Example Schematic

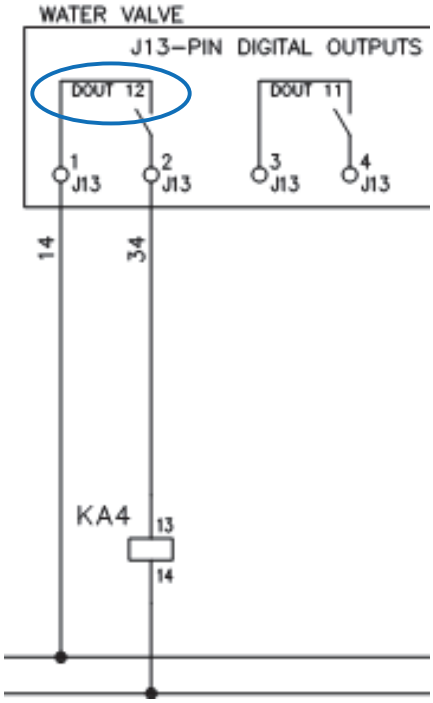


Figure 37: Example Schematic

The control relay for the starter is shown above. It is engaged anytime the compressor is running. The channel is identified as DO-16. Use the table on page 91 to find the address for digital output 16. The table shows digital output 16 as register 00202.

The transmitted message to the controller is for the 30007 register is:
01 01 00 C9 00 01 2D F4

01 Modbus address of device. This address is configured on the screen.
01 Modbus function code for reading output coils.
00 C9 Address C9 in hex is 201 in decimal. This is the relative address. The relative address is what is sent to the controller. The absolute address is used by systems such as Wonderware. The absolute address allows Wonderware to automatically determine the function code to use.
00 01 Total number of registers to get data
2D F4 CRC. This value is calculated from the message. It is used to verify messages are correct. The message is ignored by the receiver if the CRC and message do not match.

The reply from the controller is:
01 01 01 00 51 88
01 Modbus address (echo of transmitted message if successful)
01 Function code for reading input registers (echo of message if successful)
01 Indicates that one byte of data follows
00 One byte of data. The value is 0.
51 88 This value is calculated from the message. It is used to verify messages are correct.
This means CR1 is not engaged.

Example 5: Write target pressure

The address for Target Pressure is found on page 93. The absolute address given on this page is 40269. This command will change this setpoint to 101.

The transmitted message to the controller for 40269 register is:
01 06 01 0C 00 65 88 1E

01 Modbus address of device. This address is configured on the screen.
06 Modbus function code for preset single registers. Function code 16 may also be used.
01 0C Address 268. This is the relative address. Address 268 is 10C in hex.
00 65 Setpoint of 101. 101 converted to hex is 65.
88 1E CRC. This value is calculated from the message. It is used to verify messages are correct. The message is ignored by the receiver if the CRC and message do not match.

The reply from the controller is:
01 06 01 0C 00 65 88 1E

The same message is returned indicating that the controller received and processed the command.
NOTE: THIS IS NOT WORKING CORRECTLY ON THE CONTROLLER AT TIME OF THIS RELEASE. THE CONTROLLER WILL ACCEPT THE CHANGE BUT WILL HAVE 00s FOR THE DATA IN THE REPONSE.

Example 6: Remote Start

The address for remote start is found on page 95. The absolute address given on this page is 00225.

The transmitted message to the controller is for the 00225 register is:
01 05 00 E0 FF 00 8D CC

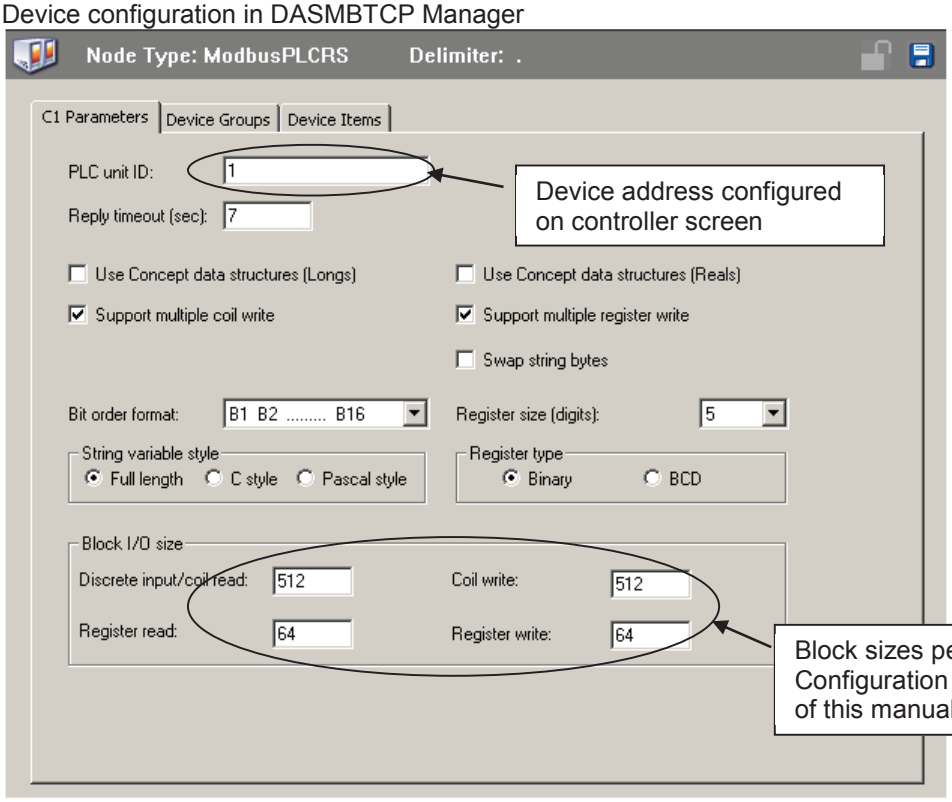
01 Modbus address of device. This address is configured on the screen.
05 Modbus function code for forcing coils. Function code 15 may also be used.
00 E0 Address 224. This is the relative address. 224 converted to hex is E0.
FF 00 FF forces the coil
8D CC CRC. This value is calculated from the message. It is used to verify messages are correct. The message is ignored by the receiver if the CRC and message do not match.

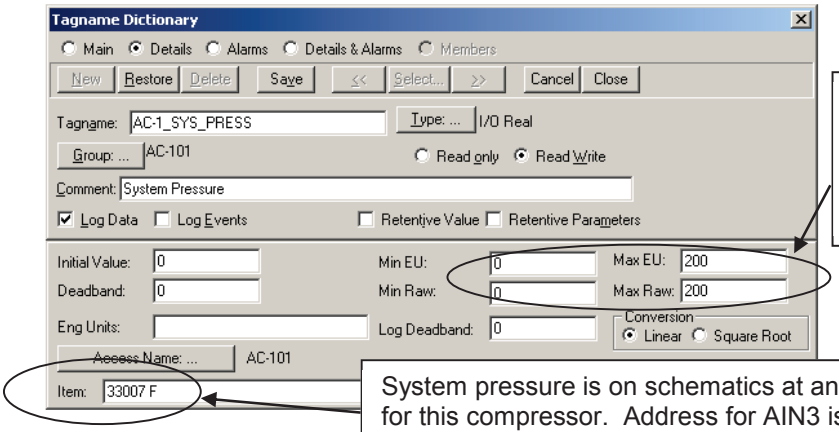
The reply from the controller is:
01 05 00 E0 FF 00 8D CC

The same message is returned indicating that the controller received and processed the command.

WONDERWARE INTOUCH

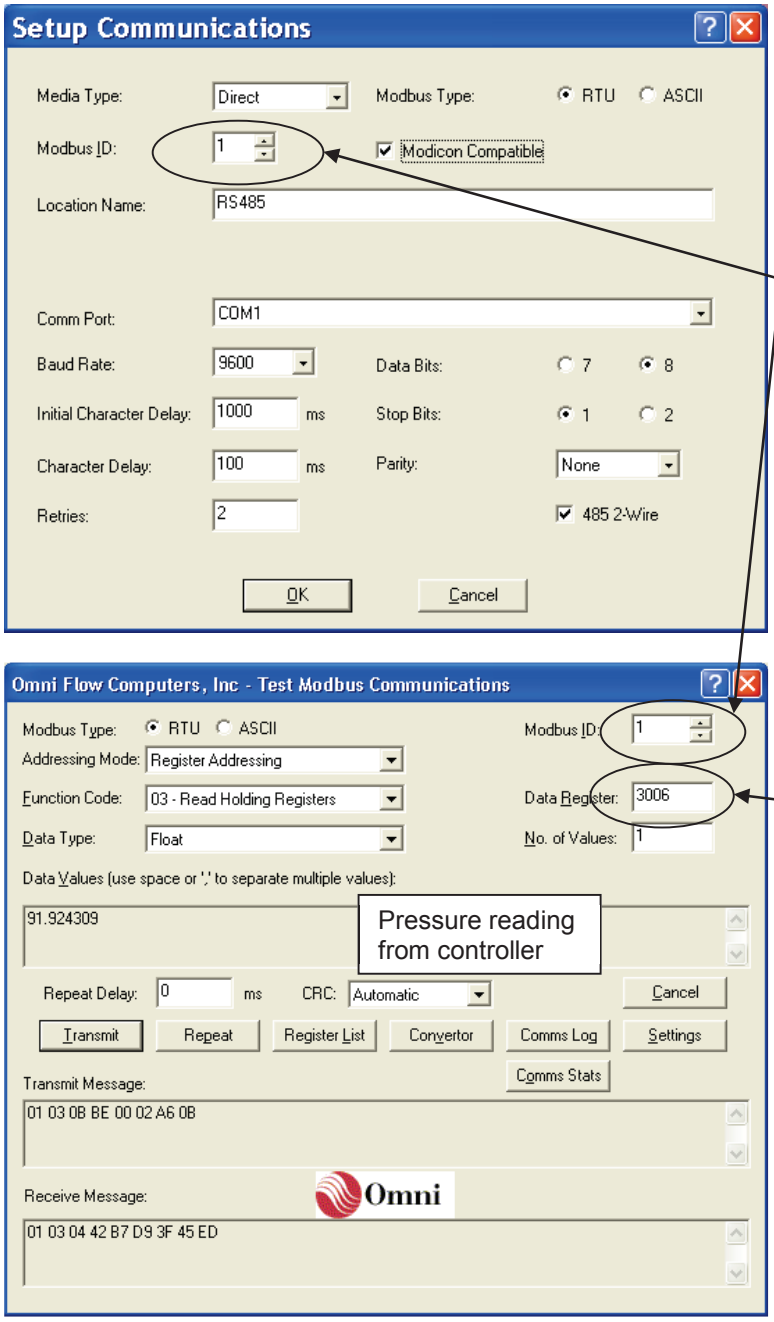
Device configuration in DASMBTCP Manager





MODBUS TESTER SOFTWARE

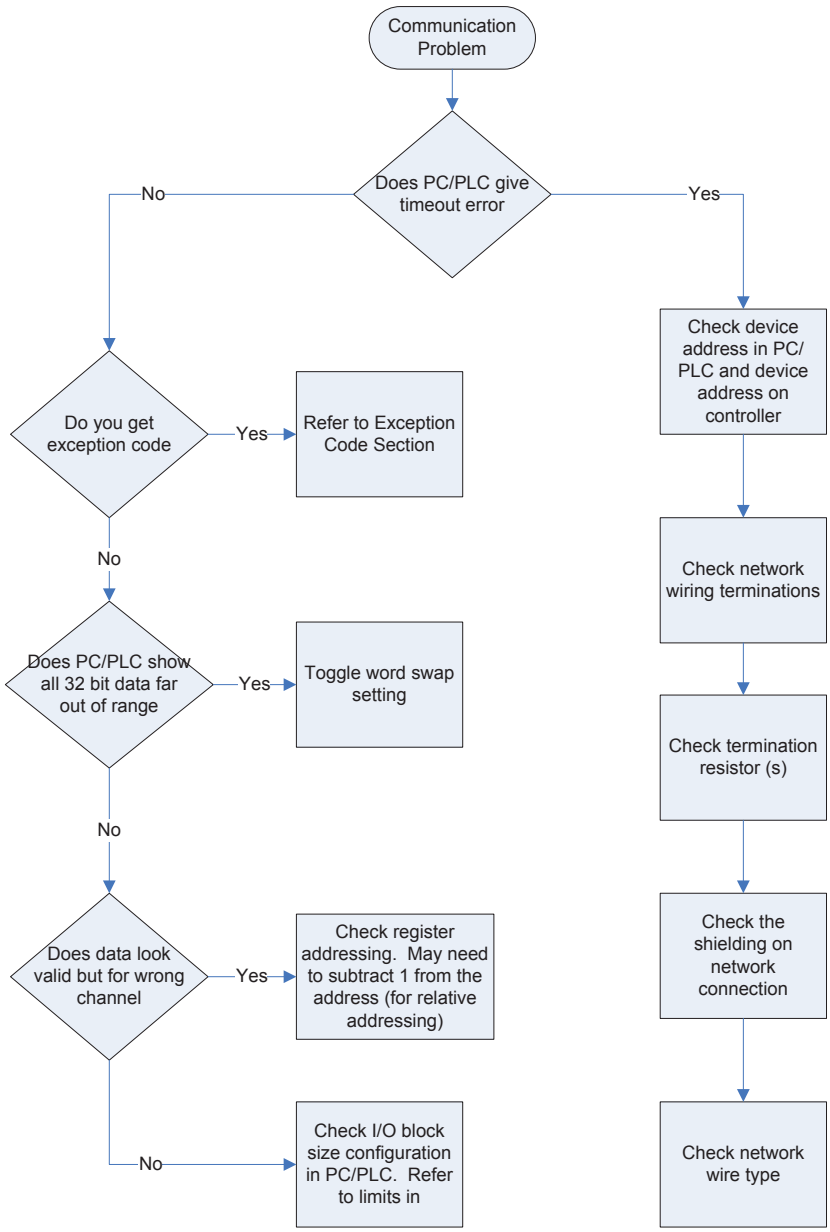
Omni Flow Computers (hiip://www omniflow com) has a great Modbus tester.



Device address configured on controller screen

Pressure reading from controller

System pressure is on schematics at analog input 3 for this compressor. Address for AIN3 is given in address section. Omni Flow with "Modicon Compatible" Addressing Mode selected uses relative addressing.


Communication Problem
Does PC/PLC give timeout error
No
Yes
Check device address in PC/ PLC and device address on controller
Do you get exception code
Yes
Refer to Exception Code Section
No
Check network wiring terminations
Does PC/PLC show all 32 bit data far out of range
Yes
Toggle word swap setting
Check termination resistor (s)
No
Check the shielding on network connection
Does data look valid but for wrong channel
Yes
Check register addressing. May need to subtract 1 from the address (for relative addressing)
Check network wire type
No
Check I/O block size configuration in PC/PLC. Refer to limits in